

Socratic AI in K–12 Science Classrooms: Effects on Critical Thinking, Motivation, and Self-Regulation in a Randomized Controlled Trial

Sean Kao

Texas A&M University

Patricia Grant

Texas A&M University

Steven Woltering

swolte@tamu.edu

Texas A&M University

Research Article

Keywords: Large language models, Socratic AI, Argumentation, Critical thinking, Self-regulation, K-12 science education, Student-AI collaboration, Randomized controlled trial

Posted Date: December 15th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-8118546/v1>

License:  This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Abstract

Teaching scientific reasoning at scale is challenging due to limited opportunities for individualized feedback. Generative AI offers a potential solution by providing adaptive scaffolding through Socratic Dialogue. In a randomized controlled trial with 90 10th-grade students, we compared three conditions: (1) Control, (2) Argument-Driven Inquiry (ADI), and (3) AI-powered ADI using ChatGPT's *Study Mode*. Students completed pre- and post-intervention assessments of scientific argumentation, critical thinking, self-efficacy, cognitive engagement, and metacognitive self-regulation. Controlling for baseline performance, students in the AI-powered condition showed significantly greater gains in scientific argumentation and critical thinking, as well as higher self-efficacy and cognitive engagement compared with both ADI and control groups. Effects on metacognitive self-regulation were nonsignificant. These findings provide the first experimental evidence that Socratic AI tutors can enhance adolescents' reasoning and engagement in real classrooms, pointing to new scalable models for supporting complex cognitive skills.

Introduction

Scientific reasoning is a core competency in K-12 science education (National Research Council, 2012; Next Generation Science Standards, 2013). It encompasses the ability to construct explanations, evaluate evidence, and revise claims through critique (Chinn & Malhotra, 2002; Sandoval, 2005; Zimmerman, 2000). Teachers, however, often face structural constraints that limit opportunities to support these skills effectively (Abate et al., 2020; Dolan & Grady, 2010; Jiménez-Aleixandre & Erduran, 2007). Students frequently struggle with the validity and structure of arguments, tending to affirm claims that align with prior beliefs, rely on confirmatory rather than disconfirmatory evidence, and inadequately sample data before drawing conclusions (Faize et al., 2017). Many also conflate observation with inference and have difficulty distinguishing between theory, evidence, and personal belief, which further complicates efforts to cultivate rigorous scientific reasoning in classroom settings (Driver et al., 2000). Even when instruction emphasizes reasoning, outcomes depend largely on the teacher's expertise, modeling ability, and adaptability during lessons (Faize et al., 2017; Schlatter et al., 2021; Shouse et al., 2007; van der Graaf et al., 2019). Artificial intelligence (AI) has emerged as a promising instructional technology capable of addressing these constraints (Su et al., 2023). AI systems can analyze student responses in real time and generate adaptive questions and feedback aligned with individual learning progress. When implemented with pedagogical intent, such systems can support learning and allow teachers to extend guidance during and beyond classroom time (Chen et al., 2023; Kestin et al., 2025). The effectiveness of AI in education depends on its design (Chen et al., 2020). AI systems that deliver information directly may promote dependency and reduce students' cognitive abilities (Abbas et al., 2024; Bastani et al., 2024; Larson et al., 2024; Melisa et al., 2025; Wu, 2024; Zhai et al., 2024). Those that guide students through Socratic questioning and explanation are poised to strengthen critical thinking skills and engagement (Dickerson, 2024). Recent research has concentrated primarily on AI as a content-generator and assessment tool (e.g., Archila et al., 2024; Awidi, 2024; Cheung et al., 2024; Oates & Johnson, 2025; Wang et al., 2024). Far fewer studies have examined AI as a Socratic tutor that engages

students in pedagogically structured scientific reasoning and critical thinking classrooms (Crompton et al., 2024); and none, to the best of our knowledge, have done so with a widely available/scalable commercial product.

This study investigates whether Socratic AI tutoring can enhance scientific reasoning and critical thinking in high school science classrooms. ChatGPT's *Study Mode* was employed to deliver adaptive Socratic Dialogue integrated within the Argument-Driven Inquiry (ADI) framework. A randomized controlled design compared three instructional conditions: AI-powered ADI, traditional ADI, and control group. The study assessed effects on scientific argumentation, critical thinking, self-efficacy, cognitive engagement, and metacognitive self-regulation.

Scientific Argumentation, Critical Thinking, and Artificial Intelligence

Scientific argumentation is a foundational practice through which students learn how scientific knowledge is constructed and evaluated (Duschl et al., 2007; Osborne et al., 2004). It requires learners to formulate claims, justify them with evidence, and assess competing explanations (Grooms et al., 2015). This process strengthens conceptual understanding by requiring students to evaluate the quality of evidence and the coherence of reasoning. It also supports the development of critical thinking, defined as the capacity to assess arguments that extend beyond surface-level structures and to make informed judgments and decisions in problem-solving contexts (Willingham, 2008). Studies have shown that explicit engagement in argumentation improves both scientific reasoning and content knowledge (von Aufschnaiter et al., 2008; Faize & Akhtar, 2020; Jimenez-Aleixandre, et al., 2000; Nussbaum & Sinatra, 2003; Sampson & Clark, 2009). In K-12 science education, ADI is a well established instructional model designed to operationalize these goals (Chen et al., 2016; Demircioglu & Ucar, 2015; Eymur, 2018; Sampson et al., 2011; Walker et al., 2012). It guides students through investigation, argument construction, peer critique, and written reflection (Walker et al., 2011). Empirical findings indicate that ADI enhances argument quality, conceptual understanding, and reasoning performance across multiple science domains (Fakhriyah et al., 2021). Its effectiveness, however, relies on consistent facilitation and timely feedback, which can be difficult to achieve in typical classrooms due to time and workload constraints (Demirtaş et al., 2025).

Generative AI technologies are poised to complement such inquiry-based models by extending feedback and scaffolding opportunities (Yeh, 2025; Zhai et al., 2021). Large language models (LLMs), such as ChatGPT, are capable of generating adaptive prompts, elicit reasoning, and guide learners through structured dialogue (Essel et al., 2024). When applied to science learning, these systems can replicate aspects of the teacher's questioning process and provide individualized guidance at scale (Khan, 2024). Early findings indicate that AI tutoring environments can improve scientific reasoning performance and engagement when learners remain active participants in the dialogue (Oppenheimer et al., 2025; Szmyd & Mitera, 2024; Tang & Putra, 2025). Socratic AI tutoring represents one implementation of this approach. It relies on iterative questioning to stimulate reflection and justification rather than direct answer provision (Gregorcic et al., 2024). The design aligns with cognitive apprenticeship and

constructivist theories, emphasizing learning through guided inquiry and metacognitive monitoring (Collins et al., 1991; Fosnot, 2013). In practice, Socratic Dialogue allows learners to test ideas, confront misconceptions, and strengthen reasoning chains through co-construction of knowledge (Mercer, 1995) in a dialogic inquiry (Wells, 1999). Recent evidence indicates that such AI-mediated questioning can enhance reasoning quality and engagement compared with traditional instruction in higher education (Essel et al., 2024; Shalong et al., 2025).

The integration of Socratic AI into scientific argument-based instruction may provide a mechanism for balancing scalability and depth of feedback (Hattie & Timperley, 2007). Teachers retain responsibility for guiding collective inquiry, while Socratic AI tutors deliver individualized questioning and explanation (Gregorcic et al., 2024). This division of labor may allow sustained engagement with scientific reasoning processes within realistic classroom constraints.

Self-Efficacy, Metacognitive Self-Regulation, and Artificial Intelligence

Self-efficacy and metacognitive self-regulation are critical determinants of students' success in critical thinking (Gurcay & Ferah, 2018; Ku & Ho, 2010; Uzuntiryaki-Kondakci & Capa-Aydin, 2013). Self-efficacy, defined as an individual's belief in their capability to perform a task effectively, includes effort, persistence, and willingness to engage with complex problems (Bandura, 1982). Learners with higher self-efficacy are more likely to sustain attention, apply learning strategies, and recover from setbacks. In science education, self-efficacy predicts not only achievement but also engagement in inquiry and argumentation tasks (Lin, 2021).

Metacognitive self-regulation refers to the ability to plan, monitor, and evaluate one's cognitive processes during learning (Brown, 1987). It allows students to identify gaps in understanding, adjust strategies, and apply feedback effectively. Together, self-efficacy and metacognitive self-regulation form the basis of self-regulated learning (SRL), a framework that explains how learners manage their motivation, cognition, and behavior to achieve learning goals (Efklides, 2011; Zimmerman, 2002). Research consistently links SRL with higher academic achievement and deeper conceptual understanding in science learning (Higgins et al., 2023; Zimmerman, 2013).

AI systems can support these processes by providing adaptive feedback and reflection prompts that help students monitor progress and calibrate self-efficacy (Jin et al., 2023; Wang et al., 2023). Studies have shown that personalized feedback from AI tutors can strengthen students' sense of competence and promote persistence on challenging tasks (Yilmaz & Yilmaz, 2023; Xu et al., 2025). Socratic questioning within AI environments also encourages metacognitive awareness by prompting learners to articulate reasoning, evaluate their responses, and plan next steps (Yilmaz Soylu et al., 2025). The effects of AI on metacognitive self-regulation depend on how control is distributed between the learner and the system (Molenaar, 2022). Excessive guidance can reduce autonomy and limit opportunities for independent regulation, whereas scaffolding that gradually transfers responsibility supports agency and

reflective learning (Boud, 2012). Adolescent learners, in particular, may respond differently to feedback due to their sensitivity to social comparison and evaluation (Dijkstra et al., 2008). AI tutors can mitigate these pressures by providing private, nonjudgmental social comparison nudging that allows students to practice reasoning and argumentative writing without direct peer evaluation (Wambsganss et al., 2022).

Current Study

The integration of Socratic AI into science education offers the potential to enhance reasoning, motivation, and self-regulation within existing instructional frameworks. However, empirical evidence on AI's classroom impact remains limited, particularly in K-12 education (Lee & Kwon, 2024). Few have examined how AI functions as a Socratic tutor in authentic K-12 environments where inquiry, collaboration, and feedback are central to instruction (Yim & Su, 2025). This study investigates the effectiveness of Socratic AI tutoring embedded within the ADI model in promoting adolescents' scientific reasoning, critical thinking, and self-regulation. ChatGPT's *Study Mode* was selected as the intervention platform because it employs Socratic questioning and reflective prompts designed to "guide understanding and promote active learning" (OpenAI, 2025). The AI was implemented to assist students during key phases of ADI (i.e., constructing arguments, evaluating evidence, preparing for rebuttals) by encouraging explanation, justification, and reflection.

A randomized controlled trial design was implemented with three conditions: (1) AI-powered ADI, where students engaged with ChatGPT's *Study Mode* during inquiry activities; (2) traditional ADI, conducted without AI support; and (3) control group, no intervention at all. This structure allowed for direct comparison of AI-powered and teacher-led inquiry while controlling for baseline differences in content exposure instructional quality.

The study assessed both cognitive and motivational outcomes. The primary outcomes were scientific argumentation and critical thinking, representing core indicators of scientific reasoning. Secondary outcomes included self-efficacy, cognitive engagement, and metacognitive self-regulation, which reflect students' motivation and self-regulatory capacities.

Three research questions guided this research:

1. Do students receiving AI-powered ADI demonstrate greater gains in scientific argumentation and critical thinking than those in traditional ADI and control conditions?
2. Do students experience higher levels of self-efficacy and cognitive engagement under Socratic AI scaffolding compared with students in traditional ADI and control conditions?
3. Do students supported by Socratic AI scaffolding report higher levels of metacognitive self-regulation than students in the traditional ADI and control conditions?

By integrating Socratic AI into a validated inquiry-based learning model, this study aims to advance understanding of how Socratic AI tutoring can enhance both cognitive and motivational dimensions of

learning. The findings are expected to inform the integration of scalable, pedagogically grounded AI systems capable of supporting scientific argument-based instruction in K-12 science education.

Method

Participants The study was conducted at a large high school in northern Taiwan with 10th-grade students aged 16–17. Participants were recruited from science classes that had previously implemented ADI as part of their curriculum, averaging 32–36 hours of prior ADI-based instruction. Eligibility criteria included: (1) enrollment in an ADI-based science course; (2) completion of an AI literacy module before the intervention; (3) participants were excluded for noncompletion); (4) ability and willingness to interact with AI tools through typed responses; and (5) completion of at least two semesters of both science (e.g., physics, chemistry, or biology) and English language courses. Participants were also required to demonstrate English proficiency, evidenced by a TOEFL Junior score ≥ 745 or equivalent. Written parental consent and student assent were obtained prior to participation (IRB approval number BLINDED). The target minimum sample size of 82 participants was determined through a power analysis based on prior meta-analytic findings indicating medium-to-large effects of collaborative, inquiry-based instruction on critical thinking (Antonio & Prudente, 2024; Xu et al., 2023). This ensured a minimum power of 0.80 for detecting significant differences among groups. Only students who completed both pre- and post-assessments were included in the analysis. A one-way analysis of variance (ANOVA) was first conducted to confirm baseline equivalence across groups. Post-intervention group differences were examined using one-way analysis of covariance (ANCOVA), with pretest scores included as covariates. Effect sizes were interpreted following Cohen's (1973) guidelines: $\eta^2 = .01-.05$ (small), $.06-.13$ (medium), and $> .14$ (large). All analyses were performed using R (version 4.5.1; R Core Team, 2025). After consent procedures, 90 eligible students were randomly assigned to one of three groups: AI-powered ADI ($n = 30$, 10 female), traditional ADI ($n = 30$, 14 female), and control ($n = 30$, 16 female). Randomization was conducted by the on-site researcher using Microsoft Excel (Version 16.0, 2024). No stratification or blocking was applied, and the random allocation sequence was concealed from school personnel responsible for participant enrollment. All participants completed the study. A chi-square test showed no significant gender differences across groups, $\chi^2(2) = 2.52$, $p = .28$. A one-way ANOVA revealed no significant difference in self-reported AI-use frequency among groups, $F(2, 87) = 2.29$, $p = .11$. Participant flow is shown in Fig. 1, following CONSORT 2025 reporting guidelines (Hopewell et al., 2025).

Materials and Procedure Data collection was conducted over two consecutive days in August 2025. All activities took place in computer laboratories under teacher supervision using school-issued computers. Day 1 began with the retrieval of parental consent forms and administration of assent forms, followed by pretesting. Participants completed three baseline assessments: the Test for Scientific Argumentation (TSA, Frey et al., 2015), the Motivated Strategies for Learning Questionnaire (MSLQ, Pintrich, 1991), and a cognitive engagement scale (Rotgans & Schmidt, 2011). Immediately following the pretest, all students completed a 30-minute online learning module introducing AI literacy and the fundamentals of scientific argumentation. The module provided an overview of how large language models such as ChatGPT function, their educational applications, and ethical considerations related to AI use. It also reviewed the

structure of scientific argumentation, including claims, evidence, reasoning, and rebuttals, drawing on Toulmin's Argument Pattern (Toulmin, 1958/2003; Simon, 2008). Day 2 consisted of the instructional intervention and posttesting. Participants were assigned to one of three conditions: (1) AI-powered ADI using ChatGPT's *Study Mode* (details provided in the next section); (2) traditional ADI; and (3) control (no intervention). The total session time was 200 minutes (i.e., four class periods). Following the intervention, participants completed the same assessments administered during pretesting to measure changes in scientific argumentation, self-efficacy, cognitive engagement, and metacognitive self-regulation. An additional essay-based writing task was administered during intervention and posttest to assess students' ability to construct extended arguments; these results are not reported in the current manuscript. Figure 2 presents the overall flow of study procedures.

After random assignment, students were placed into separate classrooms according to their assigned condition. Each classroom accommodated up to 40 students and was supervised by trained teachers to ensure procedural consistency and a controlled learning environment. Students in the control group ($n = 30$) did not receive any instructional intervention. After completing the pretest, they resumed their regular coursework, which included mathematics and English, without participation in any AI or argumentation activities. On Day 2, they returned to the computer lab to complete the posttests concurrently with the intervention groups. Students in the traditional ADI group ($n = 30$) completed a standard ADI task without AI assistance. Each participant addressed the same socio-scientific issue and composed an initial written argument by formulating a claim, collecting evidence, and providing reasoning (Braund et al., 2007). They then engaged in a double-blind peer review, offering feedback on a peer's argument. Peer pairings were randomly pre-assigned by the teacher. After receiving peer feedback, students revised and resubmitted their final reports. The instructor's role was limited to procedural facilitation and ensuring adherence to the peer review process; no direct content feedback was provided. This format was consistent with the ADI method previously used in the students' science classes. Students in the AI-powered ADI group ($n = 30$) completed the same ADI task supported by ChatGPT's *Study Mode* (GPT-5, August 2025). Sessions were held in a computer laboratory under the supervision of a classroom teacher and the on-site researcher. Each student worked individually with ChatGPT's *Study Mode* configured to deliver structured Socratic Dialogue rather than direct answers. The AI guided students through each phase of argument construction by prompting clarification, justification, and reflection. Students first developed a preliminary argument and then refined it using AI feedback. All chat logs were recorded for later analysis (see Fig. 3 for an anonymized example).

Scientific argumentation ability was assessed using the TSA, a 36-item multiple-choice test evaluating students' ability to construct and evaluate arguments. The TSA measures six components: distinguishing claims from data and evidence, identifying qualifiers, recognizing reasoning types, evaluating rebuttals, identifying counterarguments, and determining whether a statement constitutes a claim. Each item was scored dichotomously (1 = correct, 0 = incorrect). Both total and subscale scores were computed, with higher scores reflecting greater proficiency in scientific reasoning. The TSA demonstrated good internal consistency ($\alpha = .82$) and moderate concurrent validity with the *Cornell Critical Thinking Test* ($r = .59, p < .001$).

Motivation and self-regulated learning were assessed using two subscales from the MSLQ: self-efficacy and metacognitive self-regulation. The self-efficacy subscale (8 items) measures students' confidence in completing academic tasks successfully (e.g., "I'm confident I can do an excellent job on assignments and tests in this course"). The metacognitive self-regulation subscale (12 items) assesses planning, monitoring, and evaluation processes during learning (e.g., "I ask myself questions to make sure I understand the material I have been studying"). Items were rated on a 7-point Likert scale (1 = *not at all true of me* to 7 = *very true of me*). Higher scores indicate greater self-efficacy and more frequent use of self-regulatory strategies. Reported internal consistencies are $\alpha = .93$ and $\alpha = .79$, respectively, and construct validity has been established through confirmatory factor analyses (Pintrich, 1991).

Cognitive engagement was measured using the 4-item Situational Cognitive Engagement Scale, which captures students' momentary engagement and effort during a learning task (e.g., "I was engaged with the topic at hand"). Items were rated on a 7-point Likert scale (1 = *not at all true of me* to 7 = *very true of me*), and mean scores were computed, with higher values indicating greater engagement and persistence. The scale demonstrates strong internal consistency (Hancock's $H = .78 - .93$) and construct validity.

Results

The present study examined the effects of ADI and AI-powered ADI training on students' scientific reasoning and critical thinking. The primary outcome measures were scientific argumentation and critical thinking, and the secondary measures were self-efficacy, cognitive engagement, and metacognitive self-regulation. Descriptive statistics for all measures are presented in Table 1. Preliminary analyses indicated no significant group differences at pretest (all $p > .05$). Specifically, there were no differences among the three groups in scientific argumentation, $F(2, 87) = 0.46$, $p = .63$; critical thinking, $F(2, 87) = 0.10$, $p = .91$; self-efficacy, $F(2, 87) = 0.25$, $p = .78$; cognitive engagement, $F(2, 87) = 0.20$, $p = .82$; or metacognitive self-regulation, $F(2, 87) = 2.12$, $p = .12$. Levene's tests for equality of variances were not significant.

Table 1

Means and Standard Deviations of Pretest and Posttest Scores by Condition. This table includes means and standard deviations (in parentheses). ADI = Argument-Driven Inquiry group; AI = AI-supported ADI group.

	Control		ADI		AI	
	Pre	Post	Pre	Post	Pre	Post
Scientific Argumentation	20.8 (4.6)	21.2 (5.2)	21.8 (4.6)	23.6 (3.8)	21.8 (4.9)	25.5 (4.6)
Critical Thinking	20.2 (4.7)	20.5 (3.2)	19.9 (5.4)	22.2 (2.9)	20.4 (4.0)	25.4 (3.2)
Self-efficacy	30.5 (10.0)	30.9 (8.8)	29.0 (11.2)	30.0 (10.2)	30.6 (8.2)	35.7 (8.9)
Cognitive Engagement	11.9 (2.6)	12.5 (2.2)	12.0 (2.4)	14.6 (3.4)	11.6 (1.4)	17.4 (3.3)
Metacognitive Self-regulation	51.6 (8.8)	49.2 (6.8)	46.9 (8.9)	47.8 (6.9)	49.7 (7.4)	51.1 (7.0)

A one-way analysis of covariance (ANCOVA) was conducted for each dependent variable, with pretest scores included as covariates. The results showed a significant overall treatment effect for scientific argumentation, $F(2, 86) = 10.63$, $p < .001$, $\eta^2 = .20$, representing a large effect size. Adjusted means indicated that students in the AI group significantly outperformed those in the control group ($p < .001$) and the ADI group ($p = .038$). The ADI group also scored higher than the control group, though the difference approached but did not reach statistical significance ($p = .066$). For critical thinking, the ANCOVA revealed a significant main effect of treatment, $F(2, 86) = 20.20$, $p < .001$, $\eta^2 = .32$, indicating a large effect. Students in the AI group performed significantly better than both the control group ($p < .001$) and the ADI group ($p < .001$), while the ADI group also scored significantly higher than the control group ($p = .037$).

The effect of treatment on self-efficacy was also significant, $F(2, 86) = 6.43$, $p = .002$, $\eta^2 = .13$, reflecting a medium-to-large effect size. Students in the AI group reported higher adjusted posttest self-efficacy compared to those in the control ($p = .007$) and ADI ($p = .009$) groups. No difference emerged between the ADI and control groups ($p = .98$). A similar pattern was observed for cognitive engagement, $F(2, 86) = 19.90$, $p < .001$, $\eta^2 = .32$, also indicating a large effect. Students in the AI group demonstrated greater engagement than both the control ($p < .001$) and ADI ($p < .001$) groups, while the ADI group scored significantly higher than the control group ($p = .008$). For metacognitive self-regulation, however, no significant effect of treatment was found, $F(2, 86) = 2.18$, $p = .12$, $\eta^2 = .05$. This suggests that the interventions did not produce measurable differences in students' metacognitive self-regulatory learning strategies across groups.

Overall, the findings support the hypothesis that AI-powered Socratic Dialogue enhances scientific argumentation, critical thinking, self-efficacy, and cognitive engagement relative to traditional ADI and

control instruction. The absence of group differences in metacognitive self-regulation indicates that short-term exposure to Socratic AI dialogue may not immediately translate into improvements in metacognitive self-regulatory strategies. Adjusted posttest means for all outcomes are depicted in Fig. 4.

Discussion

The study examined the impact of Socratic AI-powered ADI on students' scientific reasoning and critical thinking relative to traditional ADI and control conditions. Results supported the hypotheses across most outcomes. The AI-assisted group showed stronger gains in scientific argumentation and critical thinking, with large effect sizes ($\eta^2 = .20$ and $.32$). Improvements also extended to motivational factors, including self-efficacy and cognitive engagement. Both intervention groups outperformed the control, but the AI-powered ADI produced the most consistent gains. Only metacognitive self-regulation showed no significant group difference.

The results demonstrate that a Socratic AI tutor can replicate and extend aspects of expert human guidance in scientific reasoning instruction. Through Socratic questioning and adaptive feedback, the AI guided learners to examine assumptions, test evidence, and evaluate counterarguments, skills essential for scientific argumentation development. This mechanism parallels the scaffolding principles long identified in intelligent tutoring systems research and supports findings that guided dialogue enhances reasoning and engagement nearly as effectively as human tutoring (VanLehn, 2011). The results align with recent evidence that LLM-based tutors can improve critical thinking when designed to emphasize questioning rather than direct answers (Oppenheimer et al., 2025; Tang & Putra, 2025). Socratic AI tutor appears particularly suited for fostering scientific reasoning and critical thinking in contexts where teacher-student dialogue is constrained by time and class size. By combining accessibility with individualized scaffolding, it allows scientific reasoning practice at scale without reliance on peer or instructor availability. These findings also reinforce the argument that generative AI tools, when aligned with inquiry-based models, can serve as effective cognitive partners rather than information providers (Min et al., 2025; Yeh, 2025).

Self-efficacy and cognitive engagement gains suggest that AI-powered Socratic questioning and immediate feedback has the potential to strengthen students' grit and perseverance to complete tasks at hand (Duckworth et al., 2007). The private, nonjudgemental nature of AI dialogue may also help adolescents engage more freely in scientific reasoning, reducing the social pressure often found in collaborative settings (Wambsganss et al., 2023). The individualized AI-mediated learning environment provided consistent challenge and reassurance, creating a tailored balance of difficulty and support that encouraged sustained effort. The absence of growth in metacognitive self-regulation suggests that while the AI scaffolded scientific reasoning effectively, it also assumed some reflective functions and efforts learners would normally perform by themselves. When feedback is continuous and directive, students may rely on the system's cues rather than independently planning or monitoring their metacognitive thought processes. Evidence from similar studies indicates that explicit metacognitive prompts and

gradual withdrawal of AI support are needed to cultivate self-efficacy in performance and judgment in science learning (Wang et al., 2021).

Limitations

The findings should be interpreted in light of several constraints. The intervention occurred in a single, relatively well-resourced high school with established digital infrastructure, and highly motivated students, limiting generalizability to settings with fewer resources. The duration of exposure was short, capturing immediate effects rather than long-term learning and transfer. Longer interventions may produce stronger or more generalizable outcomes. Measures of self-efficacy and engagement relied on self-report instruments that are vulnerable to bias. The AI also lacked a formal student model and relied on heuristic prompting, which may have limited adaptivity (Shute, 1995). Future studies may further examine the effectiveness of customizable Socratic AI systems capable of adapting prompts, feedback, and pacing to different learners and instructional approaches (Collins et al., 2024; Tang & Putra, 2025). Broader trials incorporating behavioral and learning analytics across varied school contexts will also be essential to evaluate the instructional value of Socratic AI in K-12 education.

Conclusion

The experiment provides early evidence that Socratic AI-powered best practices can enhance secondary students' scientific reasoning and motivation in science education. AI systems designed around guided Socratic Dialogue and adaptive feedback can complement inquiry-based and argument-based best practices by providing individualized cognitive and emotional support. The absence of improvement in metacognitive self-regulation suggests that when scaffolding is fully automated, learners may rely on the system rather than engage in independent reflection (Wu, 2024). Future research should also extend the intervention period, examine cross-domain transfer of reasoning skills (National Academy of Education, 2021; National Council for the Social Studies, 2010, 2013), and identify which elements of Socratic AI dialogue (e.g., prompt type, feedback timing, or adaptive difficulty) most influence learning. Furthermore, future studies can examine the mediating role of AI self-efficacy between AI competency and engagement in science learning. Students who understand how AI systems function may develop a higher tendency in utilizing them in learning, which in turn can increase engagement and academic performance (Ottenbreit-Leftwich et al., 2023). Investigating this mediation model would clarify the psychological mechanisms through which AI self-efficacy and competency translates into active learning and academic engagement.

Declarations

Author Contribution

S.K. and S.W. designed the intervention and assessments. S.K. ran the study. S.K. and P.G. conducted the analysis and prepared the tables and figures. S.K. conducted the literature review. All authors contributed

to the writing and editing of the manuscript.

References

1. Abate, D. (2020). Assessment of scientific reasoning: Development and validation of scientific reasoning assessment. *Eurasia Journal of Mathematics Science and Technology Education*, 16(7), em1857. <https://doi.org/10.29333/ejmste/9353>
2. Abbas, M., Jam, F. A., & Khan, T. I. (2024). Is it harmful or helpful? Examining the causes and consequences of generative AI usage among university students. *International Journal of Educational Technology in Higher Education*, 21(1), 10. <https://doi.org/10.1186/s41239-024-00444-7>
3. Antonio, R. P., & Prudente, M. S. (2024). Effects of inquiry-based approaches on students' higher-order thinking skills in Science: A meta-analysis. *International Journal of Education in Mathematics Science and Technology*, 12(1), 251–281. <https://doi.org/10.46328/ijemst.3216>
4. Archila, P. A., Ortiz, B. T., Truscott de Mejía, A. M., & Molina, J. (2024). Thinking critically about scientific information generated by ChatGPT. *Information and Learning Sciences*, 125(11/12), 1074–1106. <https://doi.org/10.1108/ILS-04-2024-0040>
5. Awidi, I. T. (2024). Comparing expert tutor evaluation of reflective essays with marking by generative artificial intelligence (AI) tool. *Computers and Education: Artificial Intelligence*, 6, 100226. <https://doi.org/10.1016/j.caeai.2024.100226>
6. Bandura, A. (1982). Self-efficacy mechanism in human agency. *American Psychologist*, 37(2), 122. <https://doi.org/10.1037/0003-066X.37.2.122>
7. Bastani, H., Bastani, O., Sungu, A., Ge, H., Kabakçı, Ö., & Mariman, R. (2024). Generative AI can harm learning. *The Wharton School Research Paper*. <https://dx.doi.org/10.2139/ssrn.4895486>
8. Boud, D. (2012). *Developing student autonomy in learning*. Routledge.
9. Braund, M., Lubben, F., Scholtz, Z., Sadeck, M., & Hodges, M. (2007). Comparing the effect of scientific and socio-scientific argumentation tasks: Lessons from South Africa. *School Science Review*.
10. Brown, A. L. (1987). Metacognition, executive control, self-regulation, and other more mysterious mechanisms. In F. Weinert, & R. Kluwe (Eds.), *Handbook of child psychology: Vol. 3. Cognitive development* (pp. 263–340). Wiley.
11. Chen, B., Zhu, X., & del Castillo, F. D. (2023). Integrating generative AI in knowledge building. *Computers and Education: Artificial Intelligence*, 5, 100184. <https://doi.org/10.1016/j.caeai.2023.100184>
12. Chen, H. T., Wang, H. H., Lu, Y. Y., Lin, H. S., & Hong, Z. R. (2016). Using a modified argument-driven inquiry to promote elementary school students' engagement in learning science and argumentation. *International Journal of Science Education*, 38(2), 170–191. <https://doi.org/10.1080/09500693.2015.1134849>

13. Chen, L., Chen, P., & Lin, Z. (2020). Artificial intelligence in education: A review. *IEEE access*, 8, 75264–75278. <https://doi.org/10.1109/ACCESS.2020.2988510>
14. Cheung, K. K. C., Pun, J. K., & Li, W. (2024). Students' holistic reading of socio-scientific texts on climate change in a ChatGPT scenario. *Research in Science Education*, 54(5), 957–976. <https://doi.org/10.1007/s11165-024-10177-2>
15. Chinn, C. A., & Malhotra, B. A. (2002). Epistemologically authentic inquiry in schools: A theoretical framework for evaluating inquiry tasks. *Science Education*, 86(2), 175–218. <https://doi.org/10.1002/sce.10001>
16. Cohen, J. (1973). Eta-squared and partial eta-squared in fixed factor ANOVA designs. *Educational and Psychological Measurement*, 33(1), 107–112. <https://doi.org/10.1177/00131644730330011>
17. Collins, A., Brown, J. S., & Holum, A. (1991). Cognitive apprenticeship: Making thinking visible. *American Educator*, 15(3), 6–11.
18. Collins, B. R., Black, E. W., & Rarey, K. E. (2024). Introducing AnatomyGPT: A customized artificial intelligence application for anatomical sciences education. *Clinical Anatomy*, 37(6), 661–669. <https://doi.org/10.1002/ca.24178>
19. Crompton, H., Jones, M. V., & Burke, D. (2024). Affordances and challenges of artificial intelligence in K-12 education: A systematic review. *Journal of Research on Technology in Education*, 56(3), 248–268. <https://doi.org/10.1080/15391523.2022.2121344>
20. Demircioglu, T., & Ucar, S. (2015). Investigating the effect of argument-driven inquiry in laboratory instruction. *Educational Sciences: Theory and Practice*, 15(1), 267–283.
21. Demirtaş, D., Yildirim, U., & Özdemir, Ö. F. (2025). A critique to argument-driven inquiry (ADI) and the development of authentic argument-driven inquiry (AADI) for optics instruction. *Physics Education*, 61(1), 015006. <https://doi.org/10.1088/1361-6552/ae1092>
22. Dickerson, P. (2024). Learning with Socrates: How generative AI and ancient pedagogy can develop students' critical thinking skills. *Artificial intelligence applications in higher education* (pp. 90–105). Routledge.
23. Dijkstra, P., Kuyper, H., Van der Werf, G., Buunk, A. P., & van der Zee, Y. G. (2008). Social comparison in the classroom: A review. *Review of Educational Research*, 78(4), 828–879. <https://doi.org/10.3102/0034654308321210>
24. Dolan, E., & Grady, J. (2010). Recognizing students' scientific reasoning: A tool for categorizing complexity of reasoning during teaching by inquiry. *Journal of science teacher education*, 21(1), 31–55. <https://doi.org/10.1007/s10972-009-9154-7>
25. Driver, R., Newton, P., & Osborne, J. (2000). Establishing the norms of scientific argumentation in classrooms. *Science Education*, 84(3), 287–312. [https://doi.org/10.1002/\(SICI\)1098-237X\(200005\)84:3%3C287::AID-SCE1%3E3.0.CO;2-A](https://doi.org/10.1002/(SICI)1098-237X(200005)84:3%3C287::AID-SCE1%3E3.0.CO;2-A)
26. Duckworth, A. L., Peterson, C., Matthews, M. D., & Kelly, D. R. (2007). Grit: Perseverance and passion for long-term goals. *Journal of Personality and Social Psychology*, 92(6), 1087. <https://doi.org/10.1037/0022-3514.92.6.1087>

27. Duelen, A., Jennes, I., & Van den Broeck, W. (2024, June). Socratic AI against disinformation: Improving critical thinking to recognize disinformation using Socratic AI. In *Proceedings of the 2024 ACM International Conference on Interactive Media Experiences* (pp. 375–381).
<https://doi.org/10.1145/3639701.3663640>
28. Duschl, R. A., Shouse, A. W., & Schweingruber, H. A. (2007). What research says about K-8 science learning and teaching. *Principal*, 87(2), 16.
29. Efklides, A. (2011). Interactions of metacognition with motivation and affect in self-regulated learning: The MASRL model. *Educational Psychologist*, 46(1), 6–25.
<https://doi.org/10.1080/00461520.2011.538645>
30. Essel, H. B., Vlachopoulos, D., Essuman, A. B., & Amankwa, J. O. (2024). ChatGPT effects on cognitive skills of undergraduate students: Receiving instant responses from AI-based conversational large language models (LLMs). *Computers and Education: Artificial Intelligence*, 6, 100198. <https://doi.org/10.1016/j.caeai.2023.100198>
31. Eymur, G. (2018). Developing high school students' self-efficacy and perceptions about inquiry and laboratory skills through argument-driven inquiry. *Journal of Chemical Education*, 95(5), 709–715.
<https://doi.org/10.1021/acs.jchemed.7b00934>
32. Faize, F. A., & Akhtar, M. (2020). Addressing environmental knowledge and environmental attitude in undergraduate students through scientific argumentation. *Journal of Cleaner Production*, 252, 119928. <https://doi.org/10.1016/j.jclepro.2019.119928>
33. Faize, F. A., Husain, W., & Nisar, F. (2017). A critical review of scientific argumentation in science education. *Eurasia Journal of Mathematics Science and Technology Education*, 14(1), 475–483.
<https://doi.org/10.12973/ejmste/80353>
34. Fakhriyah, F., Rusilowati, A., & Susilaningsih, E. (2021). Argument-driven inquiry learning model: A systematic review. *International Journal of Research in Education and Science*, 7(3), 767–784.
<https://doi.org/10.46328/ijres.2001>
35. Fosnot, C. T. (2013). *Constructivism: Theory, perspectives, and practice*. Teachers College.
36. Frey, B. B., Ellis, J. D., Bulgreen, J. A., Hare, J. C., & Ault, M. (2015). Development of a test of scientific argumentation. *Electronic Journal of Science Education*, 19(4), n4.
37. Gregorcic, B., Polverini, G., & Sarlah, A. (2024). ChatGPT as a tool for honing teachers' Socratic dialogue skills. *Physics education*, 59(4), 045005. <https://doi.org/10.1088/1361-6552/ad3d21>
38. Grooms, J., Enderle, P., & Sampson, V. (2015). Coordinating scientific argumentation and the next generation Science standards through argument-driven inquiry. *Science Educator*, 24(1), 45–50.
39. Gurcay, D., & Ferah, H. O. (2018). High school students' critical thinking related to their metacognitive self-regulation and Physics self-efficacy beliefs. *Journal of Education and Training Studies*, 6(4), 125–130. <https://doi.org/10.11114/jets.v6i4.2980>
40. Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research*, 77(1), 81–112. <https://doi.org/10.3102/003465430298487>

41. Higgins, N. L., Rathner, J. A., & Frankland, S. (2023). Development of self-regulated learning: a longitudinal study on academic performance in undergraduate science. *Research in Science & Technological Education*, 41(4), 1242–1266. <https://doi.org/10.1080/02635143.2021.1997978>
42. Hopewell, S., Chan, A. W., Collins, G. S., Hróbjartsson, A., Moher, D., Schulz, K. F., & Boutron, I. (2025). CONSORT 2025 statement: Updated guideline for reporting randomised trials. *The Lancet*, 405(10489), 1633–1640. [https://doi.org/10.1016/S0140-6736\(25\)00672-5](https://doi.org/10.1016/S0140-6736(25)00672-5)
43. Jiménez-Aleixandre, M. P., & Erduran, S. (2007). Argumentation in science education: An overview. *Argumentation in science education: Perspectives from classroom-based research*, 3–27. https://doi.org/10.1007/978-1-4020-6670-2_1
44. Jiménez-Aleixandre, M. P., Bugallo Rodríguez, A., & Duschl, R. A. (2000). Doing the lesson or doing science: Argument in high school genetics. *Science Education*, 84(6), 757–792. [https://doi.org/10.1002/1098-237X\(200011\)84:6%3C757::AID-SCE5%3E3.0.CO;2-F](https://doi.org/10.1002/1098-237X(200011)84:6%3C757::AID-SCE5%3E3.0.CO;2-F)
45. Jin, S. H., Im, K., Yoo, M., Roll, I., & Seo, K. (2023). Supporting students' self-regulated learning in online learning using artificial intelligence applications. *International Journal of Educational Technology in Higher Education*, 20(1), 37. <https://doi.org/10.1186/s41239-023-00406-5>
46. Kestin, G., Miller, K., Klales, A., Milbourne, T., & Ponti, G. (2025). AI tutoring outperforms in-class active learning: an RCT introducing a novel research-based design in an authentic educational setting. *Scientific Reports*, 15(1), 17458. <https://doi.org/10.1038/s41598-025-97652-6>
47. Khan, S. (2024). *Brave new worlds: How AI will revolutionize education (and why that's a good thing)*. Viking.
48. Ku, K. Y. L., & Ho, I. T. (2010). Metacognitive strategies that enhance critical thinking. *Metacognition and Learning*, 5(3), 251–267. <https://doi.org/10.1007/s11409-010-9060-6>
49. Larson, B. Z., Moser, C., Caza, A., Muehlfeld, K., & Colombo, L. A. (2024). Critical thinking in the age of generative AI. *Academy of Management Learning & Education*, 23(3), 373–378. <https://doi.org/10.5465/amle.2024.0338>
50. Lee, S. J., & Kwon, K. (2024). A systematic review of AI education in K-12 classrooms from 2018 to 2023: Topics, strategies, and learning outcomes. *Computers and Education: Artificial Intelligence*, 6, 100211. <https://doi.org/10.1016/j.caeai.2024.100211>
51. Lin, T. J. (2021). Multi-dimensional explorations into the relationships between high school students' science learning self-efficacy and engagement. *International Journal of Science Education*, 43(8), 1193–1207. <https://doi.org/10.1080/09500693.2021.1904523>
52. Melisa, R., Ashadi, A., Triastuti, A., Hidayati, S., Salido, A., Ero, P. E. L., & Al Fuad, Z. (2025). Critical thinking in the age of AI: A systematic review of AI's effects on higher education. *Educational Process: International Journal*, 14, e2025031. <https://doi.org/10.22521/edupij.2025.14.31>
53. Mercer, N. (1995). *The guided construction of knowledge: Talk amongst teachers and learners*. Multilingual matters.
54. Microsoft Corporation (2024). *Microsoft Excel* (Version 16.0) [Computer software]. Microsoft. <https://www.microsoft.com/>

55. Min, T., Lee, B., & Jho, H. (2025). Integrating generative artificial intelligence in the design of scientific inquiry for middle school students. *Education and Information Technologies*, 1–32. <https://doi.org/10.1007/s10639-025-13410-1>
56. Molenaar, I. (2022). The concept of hybrid human-AI regulation: Exemplifying how to support young learners' self-regulated learning. *Computers and Education: Artificial Intelligence*, 3, 100070. <https://doi.org/10.1016/j.caeai.2022.100070>
57. National Council for the Social Studies College Career, and Civic life framework for social studies state standards 2013 Author Silver Spring, MD.
58. National Council for the Social Studies National Curriculum standards for social studies 2010 Author Baltimore, MD.
59. National Research Council, Division of Behavioral, Social Sciences, Board on Science Education, & Committee on a Conceptual Framework for New K-12 Science Education Standards. (2012). *A framework for K-12 science education: Practices, crosscutting concepts, and core ideas*. National Academies.
60. NGSS Lead States. (2013). *Next generation science standards: For states, by states*. National Academy.
61. Nussbaum, E. M., & Sinatra, G. M. (2003). Argument and conceptual engagement. *Contemporary Educational Psychology*, 28(3), 384–395. [https://doi.org/10.1016/S0361-476X\(02\)00038-3](https://doi.org/10.1016/S0361-476X(02)00038-3)
62. Oates, A., & Johnson, D. (2025). ChatGPT in the classroom: Evaluating its role in fostering critical evaluation skills. *International Journal of Artificial Intelligence in Education*, 1–32. <https://doi.org/10.1007/s40593-024-00452-8>
63. OpenAI (2025). *Introducing study mode – A new way to learn in ChatGPT*. Jul. 29.
64. Oppenheimer, D. M., Cash, T. N., & Pensky, A. E. C. (2025). You've got AI friend in me: LLMs as collaborative learning partners. *International Journal of Artificial Intelligence in Education*, 1–26. <https://doi.org/10.1007/s40593-025-00521-6>
65. Osborne, J., Erduran, S., & Simon, S. (2004). Enhancing the quality of argumentation in school science. *Journal of research in science teaching*, 41(10), 994–1020. <https://doi.org/10.1002/tea.20035>
66. Ottenbreit-Leftwich, A., Glazewski, K., Jeon, M., Jantaraweragul, K., Hmelo-Silver, C. E., Scribner, A., & Lester, J. (2023). Lessons learned for AI education with elementary students and teachers. *International Journal of Artificial Intelligence in Education*, 33(2), 267–289. <https://doi.org/10.1007/s40593-022-00304-3>
67. Pintrich, P. R. (1991). A manual for the use of the Motivated Strategies for Learning Questionnaire (MSLQ).
68. R Core Team (2025). *R: A language and environment for statistical computing* (version 4.5.1) [Computer software]. R Foundation for Statistical Computing. <https://www.r-project.org/>
69. Rotgans, J. I., & Schmidt, H. G. (2011). Cognitive engagement in the problem-based learning classroom. *Advances in health sciences education*, 16, 465–479. <https://doi.org/10.1007/s10459->

70. Sampson, V., & Clark, D. (2009). The impact of collaboration on the outcomes of scientific argumentation. *Science education*, 93(3), 448–484. <https://doi.org/10.1002/sce.20306>
71. Sampson, V., Grooms, J., & Walker, J. P. (2011). Argument-Driven Inquiry as a way to help students learn how to participate in scientific argumentation and craft written arguments: An exploratory study. *Science Education*, 95(2), 217–257. <https://doi.org/10.1002/sce.20421>
72. Sandoval, W. A. (2005). Understanding students' practical epistemologies and their influence on learning through inquiry. *Science Education*, 89(4), 634–656. <https://doi.org/10.1002/sce.20065>
73. Schlatter, E., Schneider, R. M., & Fischer, F. (2021). Learning scientific reasoning: A latent transition analysis. *Learning and Instruction*, 74, 101460. <https://doi.org/10.1016/j.lindif.2021.102043>
74. Shalong, W., Yi, Z., Bin, Z., Ganglei, L., Jinyu, Z., Yanwen, Z., & Feng, R. (2025). Enhancing self-directed learning with custom GPT AI facilitation among medical students: A randomized controlled trial. *Medical Teacher*, 47(7), 1126–1133. <https://doi.org/10.1080/0142159x.2024.2413023>
75. Shouse, A. W., Schweingruber, H. A., & Duschl, R. A. (Eds.). (2007). *Taking science to school: Learning and teaching science in grades K-8*. National Academies.
76. Shute, V. J. (1995). SMART: Student modeling approach for responsive tutoring. *User Modeling and User-Adapted Interaction*, 5(1), 1–44. <https://doi.org/10.1007/BF01101800>
77. Simon, S. (2008). Using Toulmin's argument pattern in the evaluation of argumentation in school science. *International Journal of Research & Method in Education*, 31(3), 277–289. <https://doi.org/10.1080/17437270802417176>
78. Su, Y., Lin, Y., & Lai, C. (2023). Collaborating with ChatGPT in argumentative writing classrooms. *Assessing Writing*, 57, 100752. <https://doi.org/10.1016/j.asw.2023.100752>
79. Szmyd, K., & Mitera, E. (2024). The impact of artificial intelligence on the development of critical thinking skills in students. *European Research Studies Journal*, 27(2), 1022–1039. <https://doi.org/10.35808/ersj/3876>
80. Tang, K. S., & Putra, G. B. S. (2025). Generative AI as a dialogic partner: Enhancing multiple perspectives, reasoning, and argumentation in science education with customized chatbots. *Journal of Science Education and Technology*, 1–13. <https://doi.org/10.1007/s10956-025-10240-1>
81. Toulmin, S. E. (1958). *The uses of argument*. Cambridge University Press.
82. Uzuntiryaki-Kondakci, E., & Capa-Aydin, Y. (2013). Predicting critical thinking skills of university students through metacognitive self-regulation skills and chemistry self-efficacy. *Educational Sciences: Theory and Practice*, 13(1), 666–670.
83. van der Graaf, J., Segers, E., & Verhoeven, L. (2019). A combined approach to strengthen children's scientific thinking: Direct instruction on scientific reasoning and training of teacher's verbal support. *International Journal of Science Education*, 41(9), 1119–1138. <https://doi.org/10.1080/09500693.2019.1594442>

84. VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist*, 46(4), 197–221. <https://doi.org/10.1080/00461520.2011.611369>
85. Von Aufschnaiter, C., Erduran, S., Osborne, J., & Simon, S. (2008). Arguing to learn and learning to argue: Case studies of how students' argumentation relates to their scientific knowledge. *Journal of Research in Science Teaching: The Official Journal of the National Association for Research in Science Teaching*, 45(1), 101–131. <https://doi.org/10.1002/tea.20213>
86. Walker, J. P., Sampson, V., & Zimmerman, C. O. (2011). Argument-driven inquiry: An introduction to a new instructional model for use in undergraduate chemistry labs. *Journal of Chemical Education*, 88(8), 1048–1056.
87. Walker, J. P., Sampson, V., Grooms, J., Anderson, B., & Zimmerman, C. O. (2012). Argument-driven inquiry in undergraduate chemistry labs: The impact on students' conceptual understanding, argument skills, and attitudes toward science. *Journal of College Science Teaching*, 41(4), 74–81.
88. Wambsganss, T., Janson, A., & Leimeister, J. M. (2022). Enhancing argumentative writing with automated feedback and social comparison nudging. *Computers & Education*, 191, 104644. <https://doi.org/10.1016/j.compedu.2022.104644>
89. Wang, H. S., Chen, S., & Yen, M. H. (2021). Effects of metacognitive scaffolding on students' performance and confidence judgments in simulation-based inquiry. *Physical Review Physics Education Research*, 17(2), 020108. <https://doi.org/10.1103/PhysRevPhysEducRes.17.020108>
90. Wang, L., Chen, X., Wang, C., Xu, L., Shadiev, R., & Li, Y. (2024). ChatGPT's capabilities in providing feedback on undergraduate students' argumentation: A case study. *Thinking Skills and Creativity*, 51, 101440. <https://doi.org/10.1016/j.tsc.2023.101440>
91. Wang, S., Sun, Z., & Chen, Y. (2023). Effects of higher education institutes' artificial intelligence capability on students' self-efficacy, creativity and learning performance. *Education and Information Technologies*, 28(5), 4919–4939. <https://doi.org/10.1007/s10639-022-11338-4>
92. Wells, C. G. (1999). *Dialogic inquiry* (Vol. 10). Cambridge University Press.
93. Willingham, D. T. (2008). Critical thinking: Why it is so hard to teach? *Arts Education Policy Review*, 109(4), 21–32. <http://dx.doi.org/10.3200/AEPR.109.4.21-32>
94. Wu, Y. (2024). Critical thinking pedagogics design in an era of ChatGPT and other AI tools—Shifting from teaching what to teaching why and how. *Journal of Education and Development*, 8(1), 1. <https://doi.org/10.20849/jed.v8i1.1404>
95. Xu, E., Wang, W., & Wang, Q. (2023). The effectiveness of collaborative problem solving in promoting students' critical thinking: A meta-analysis based on empirical literature. *Humanities and Social Sciences Communications*, 10(1), 1–11. <https://doi.org/10.1057/s41599-023-01508-1>
96. Xu, Q., Liu, Y., & Li, X. (2025). Unlocking student potential: How AI-driven personalized feedback shapes goal achievement, self-efficacy, and learning engagement through a self-determination lens. *Learning and Motivation*, 91, 102138. <https://doi.org/10.1016/j.lmot.2025.102138>

97. Xu, X., Qiao, L., Cheng, N., Liu, H., & Zhao, W. (2025). Enhancing self-regulated learning and learning experience in generative AI environments: The critical role of metacognitive support. *British Journal of Educational Technology*. <https://doi.org/10.1111/bjet.13599>
98. Yeh, H. C. (2025). The synergy of generative AI and inquiry-based learning: Transforming the landscape of English teaching and learning. *Interactive Learning Environments*, 33(1), 88–102. <https://doi.org/10.1080/10494820.2024.2335491>
99. Yilmaz Soylu, M., Lee, J., Hung, J. T., Cui, C. Z., & Joyner, D. (2025). AI literacy as a key driver of user experience in AI-powered assessment: Insights from Socratic mind. *Interactive Learning Environments*, 1–17. <https://doi.org/10.1080/10494820.2025.2564739>
100. Yilmaz, R., & Yilmaz, F. G. K. (2023). The effect of generative artificial intelligence (AI)-based tool use on students' computational thinking skills, programming self-efficacy and motivation. *Computers and Education: Artificial Intelligence*, 4, 100147. <https://doi.org/10.1016/j.caeai.2023.100147>
101. Yim, I. H. Y., & Su, J. (2025). Artificial intelligence (AI) learning tools in K-12 education: A scoping review. *Journal of Computers in Education*, 12(1), 93–131. <https://doi.org/10.1007/s40692-023-00304-9>
102. Zhai, C., Wibowo, S., & Li, L. D. (2024). The effects of over-reliance on AI dialogue systems on students' cognitive abilities: A systematic review. *Smart Learning Environments*, 11(1), 28. <https://doi.org/10.1186/s40561-024-00316-7>
103. Zhai, X., Chu, X., Chai, C. S., Jong, M. S. Y., Istenic, A., Spector, M., & Li, Y. (2021). A review of artificial intelligence (AI) in education from 2010 to 2020. *Complexity*, 2021(1), 8812542. <https://doi.org/10.1155/2021/8812542>
104. Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. *Theory into Practice*, 41(2), 64–70. https://doi.org/10.1207/s15430421tip4102_2
105. Zimmerman, B. J. (2013). Theories of self-regulated learning and academic achievement: An overview and analysis. In B. J. Zimmerman, & D. H. Schunk (Eds.), *Self-regulated learning and academic achievement* (2nd ed., pp. 1–36). Routledge.
106. Zimmerman, C. (2000). The development of scientific reasoning skills. *Developmental Review*, 20(1), 99–149. <https://doi.org/10.1006/drev.1999.049>

Figures

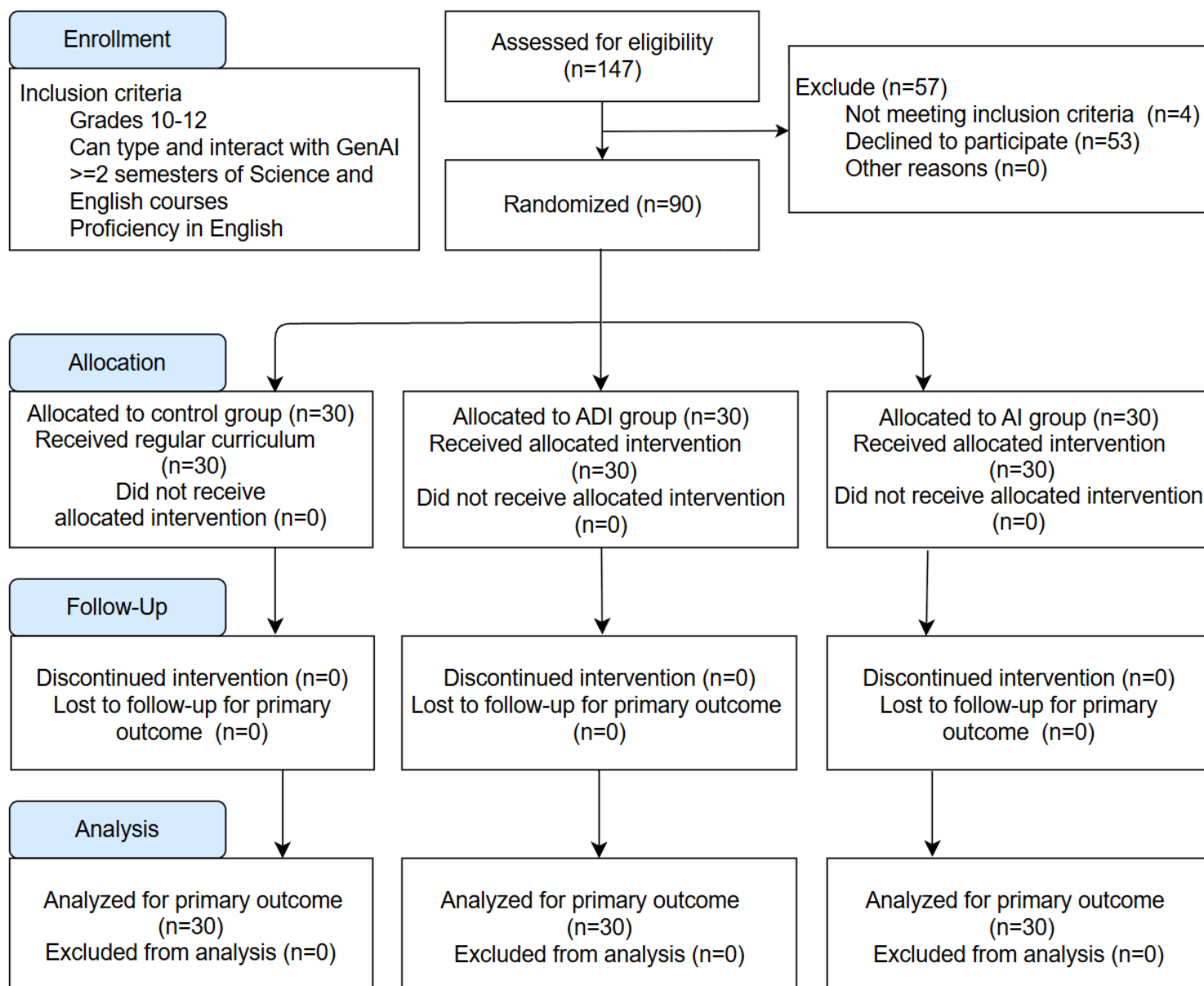


Figure 1

CONSORT Diagram.Flow diagram of participant enrollment, randomization, group allocation, follow-up, and analysis for the Control, Argument-Driven Inquiry (ADI), and AI-supported ADI (AI) groups.

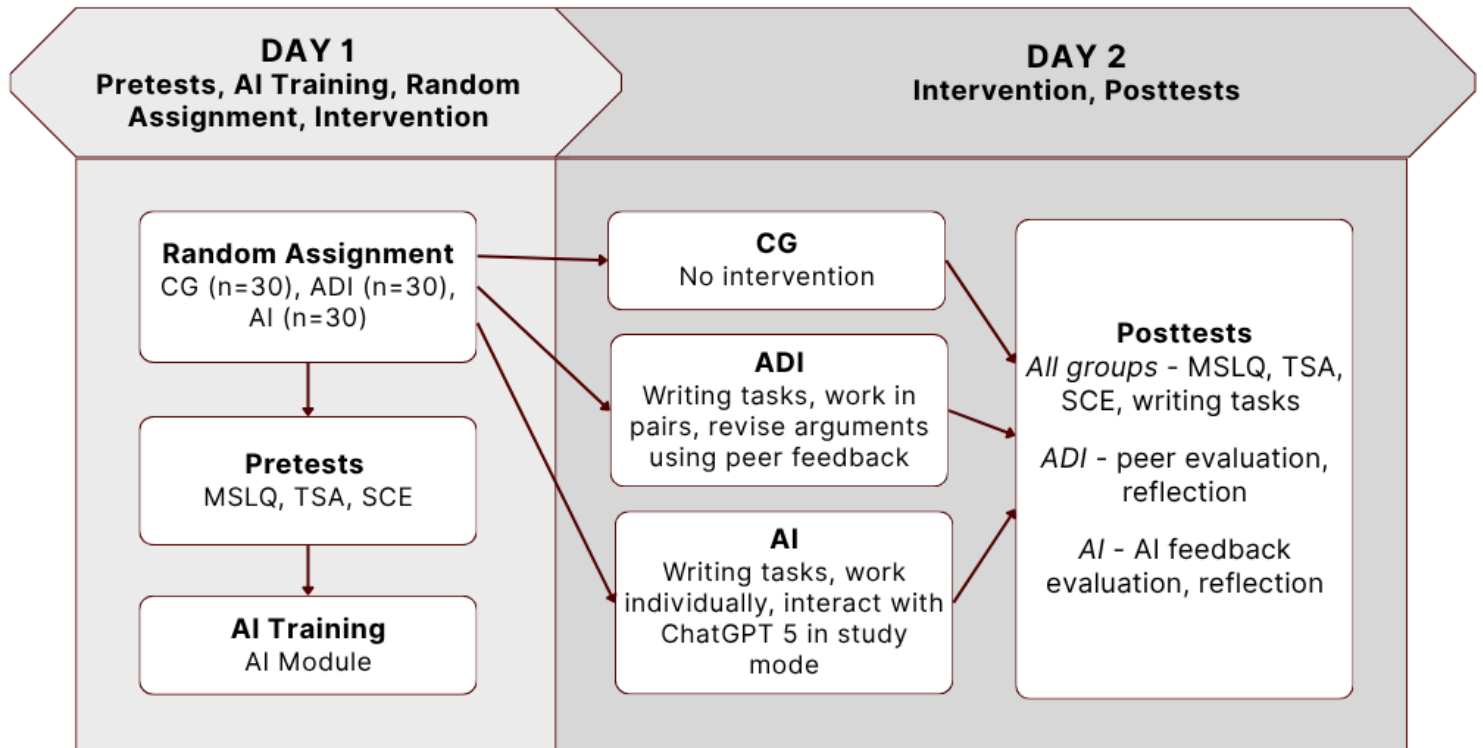


Figure 2

Study Flowchart. Day 1 involved random assignment, pretesting, and AI training. Day 2 included interventions, posttesting (identical to pretest), and group-specific evaluations (ADI: peer evaluation/reflection; AI: AI feedback/reflection).

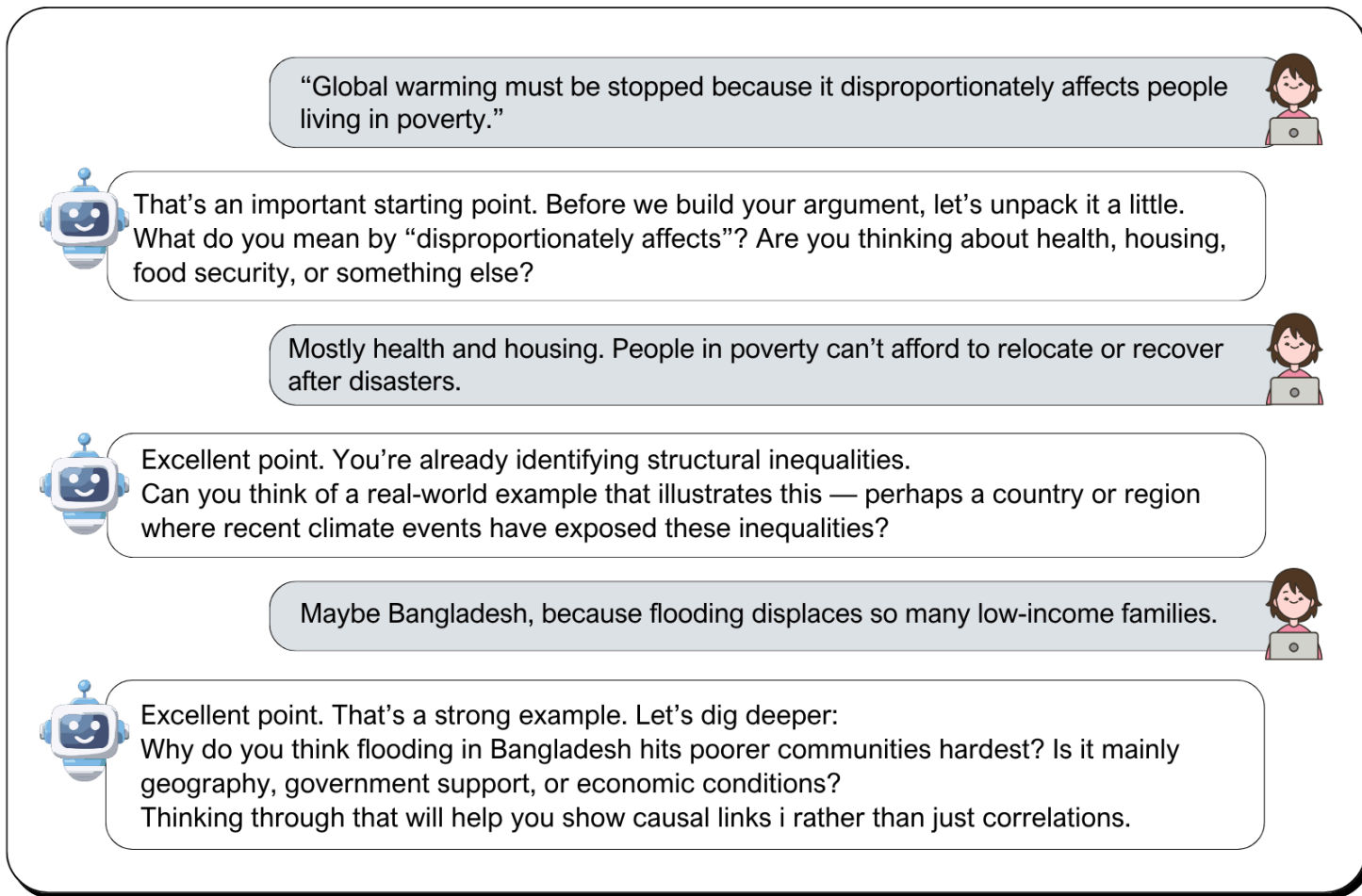


Figure 3

Example Interaction Between Participant and ChatGPT in Study Mode. This figure illustrates ChatGPT’s use of Socratic questioning to support the learner in strengthening socio-scientific argumentation.

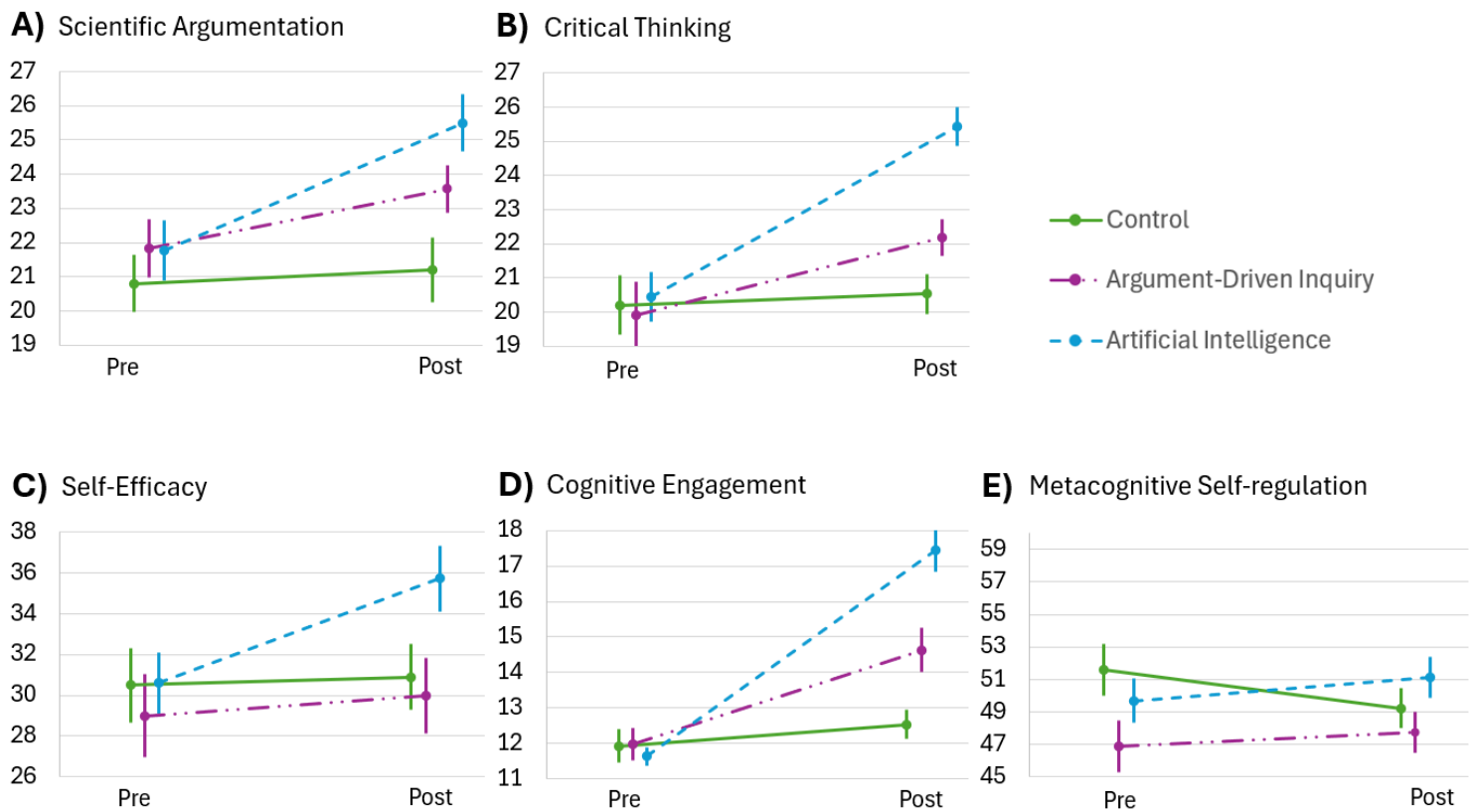


Figure 4

Mean Pre- and Post-Treatment by Condition for All Outcome Measures. This figure depicts the mean pre- and post-intervention scores for three instructional groups, Control, Argument-Driven Inquiry, and Artificial Intelligence, across five dependent variables: A) Scientific Argumentation, B) Critical Thinking, C) Self-Efficacy, D) Cognitive Engagement, and E) Metacognitive Self-regulation. Error bars represent the Standard Error of the Mean (SEM).